
Simulating Glycemic Responses to South American Desserts

WITH THE BERGMAN MINIMAL MODEL

ENG-270 Project

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1 Deviations from project proposal

Despite no major changes being made, there is one planned feature unable to be created. Compared with the original proposal, the final implementation drops the planned automatic barcode-based import from OpenFoodFacts in favour of frozen YAML snapshots verified against physical packaging (on OpenFoodFacts) and replaces the proposed analytic single-compartment verification and explicit reporting of appearance parameters ($A_{\text{fast}}, A_{\text{slow}}, k_{\text{fast}}, k_{\text{slow}}$) with a simpler numerical test suite (i.e. fasting steady state plus dt -halving peak convergence), whilst keeping the same dessert set, 240 min/1440 min simulation horizons, and control metrics (peak glucose rise ΔG , time to peak, iAUC_{0-120} , iAUC_{0-240} , time to baseline, and insulin peak time) as outlined in the proposal document in my previous submission.

2 Introduction to the problem

Postprandial (post-meal) glycemia is a critical physiological parameter reflecting the body's ability to regulate blood glucose levels after a meal; dysregulation of this process is a hallmark of metabolic disorders such as diabetes.[9, 13] Therefore, understanding how different macronutrient compositions affect glycemic appearance is essential for dietary management and prevention of metabolic diseases.[5] This project's in-baked "meals" are desserts only, as those tend to cause the highest glycemic uptake due to, at times, outrageously high sugar content, with little to no surrounding macronutrients to provide a 'cushioning' effect.

The Bergman minimal model [6, 7] provides a mathematical framework to describe this interaction between glucose and insulin. Through the simplification of a complex physiological system into a minimal set of 'compartments' and parameters, it can effectively reproduce observed dynamics.

This project aims to compare the glycemic impact of several popular south-american desserts (Açaí Bowl[1], Alfajor[2], Brigadeiro[3], Chocotorta[4]) on a nominal adult subject. South american desserts are unique in their abnormally highly sugary palette, pushing this model to its limits. A simulation pipeline that maps standard nutrition labels is utilised to model parameters to predict postprandial glucose and insulin trajectories.

3 Approach

3.1 Bergman minimal model

The core of this simulation is the Bergman minimal model. Developed by Richard Bergman and Claudio Cobelli in 1979, this tried-and-tested model has served as the rigorous foundation of glucose homeostasis research for over forty years, and is described by the following system of ordinary differential equations (ODEs): [7]

$$\frac{dG}{dt} = -(S_G + X)(G - G_b) + D(t), \quad (1)$$

$$\frac{dX}{dt} = -p_2 X + p_3(I - I_b), \quad (2)$$

$$\frac{dI}{dt} = -n(I - I_b) + u(t), \quad (3)$$

where $G(t)$ is the plasma glucose concentration, $I(t)$ is the plasma insulin concentration, and $X(t)$ represents the insulin action in a remote compartment. [6] G_b and I_b are basal values (resting state values). $D(t)$ represents the rate of glucose appearance from the gut, and, lastly, $u(t)$ represents the protein-mediated insulin secretion stimulus.

This whole system is integrated using a C backend implementing the classical explicit Runge–Kutta 4 (RK4) method to ensure numerical stability and performance. The RK4 method is a widely used numerical integration technique that advances the simulation by calculating a weighted average of

four slope estimates—determined at the start, midpoint, and end of each time step—to achieve very high accuracy with a fourth-order error convergence (i.e. it’s accurate to the fourth degree) These C functions are then exposed to the Python pipeline via `ctypes`, so all simulations can be orchestrated from a single command-line interface, making it as easy as possible for clients to work with.

It is worth noting that, while the current architecture invokes the C-kernel iteratively (introducing Python overhead), this design was explicitly chosen to prioritise the readability of the logic flow and modularity of the Python orchestrator over raw execution speed, which remains negligible for single-person simulations.

3.2 Nutrition-label mapping

To simulate real-world food items, macronutrient content from nutrition labels (carbohydrates, sugars, fiber, fat, protein) are mapped to the model’s appearance inputs.

The available carbohydrate C_{avail} is then calculated by subtracting a fraction of the fiber content, reflecting its partial indigestibility: [14]

$$C_{\text{avail}} = \max(0, \text{carbs}_g - 0.5 \cdot \text{fiber}_g).$$

The glucose appearance is modelled as a sum of two exponential decays (fast and slow pools); the fraction of carbohydrates assigned to the fast pool is determined by the sugar content. The appearance rate constants and the effective bioavailability fraction f_{app} are modulated by the presence of fat and fiber, which are known to delay gastric emptying and absorption, further highlighting the appearance of differing foods with varying macronutrients. [8, 11, 12]

Moreover, protein content drives an additional insulinotropic (affecting insulin) effect, modelled as an exponential stimulus

$$u(t) = A_{\text{prot}} e^{-k_{\text{prot}} t},$$

acknowledging amino acids’ ability to stimulate insulin secretion independently of glucose. [10]

All dessert definitions are stored as “frozen” YAML files under `configs/frozen/`, each encoding a single portion’s macronutrient content (in grams). These frozen inputs make simulations easy to reproduce and simple to extend: clients can add new desserts by copying and editing one of these files.

On another note, various numerical sanity checks, such as verifying steady-state maintenance in the absence of a meal (verifying baseline conditions remain if a ‘ghost-meal’ with no macros is input) and testing for convergence with time-step halving, were performed to validate the solver (these can even be tested independently by the client by running: `python -m pytest -q`).

3.3 Implementation and reproducibility

This project is implemented as a single Python command-line interface, with `run.py` orchestrating all simulations along with output generation. Model parameters are stored in `configs/params.yaml` with unit-encoded keys for clarity, whilst dessert-specific inputs reside in `configs/frozen/`. Running `python run.py -reproduce` replays all frozen desserts, writes zoomed and full-day figures to newly made folders `figures/zoom/` and `figures/full/`, all the while generating the summary table `tables/summary.csv`. Moreover, an additional `python run.py -reproduce -sensitivity` performs $\pm 20\%$ sweeps of fat, fiber, and protein for each dessert and records the resulting metrics in `tables/sensitivity.csv`.

Furthermore, to reassure the client’s engineering team about numerical robustness, the repository includes a minimal, but targeted, test suite in `tests/test_sanity.py`. One test checks that a “no-meal” simulation remains close to fasting steady state (maintaining basal levels within a tolerance of $< 10^{-6}$ mg/dL over 24 hours), whilst another verifies that halving the integration time step leaves the peak glucose excursion nearly unchanged. For the sake of efficiency, both tests execute with `python -m pytest -q` and currently pass, indicating this project’s RK4 implementation and Python–C interface behave consistently across runs and machines.

4 Results

The simulation was run for the four aforementioned desserts; the results are below.

4.1 Dessert dynamics

Figures 3a through 3d (see Appendix) show the zoomed view (0 min to 240 min) of the glucose and insulin response for each dessert. Figure 2 displays the full 24-hour simulation for the Açaí Bowl, demonstrating the return to basal levels. To provide a consolidated view of these disparate profiles, Figure 1 illustrates the key metrics side-by-side, distinguishing magnitude (Peak G) from total exposure (iAUC) and temporal delays.

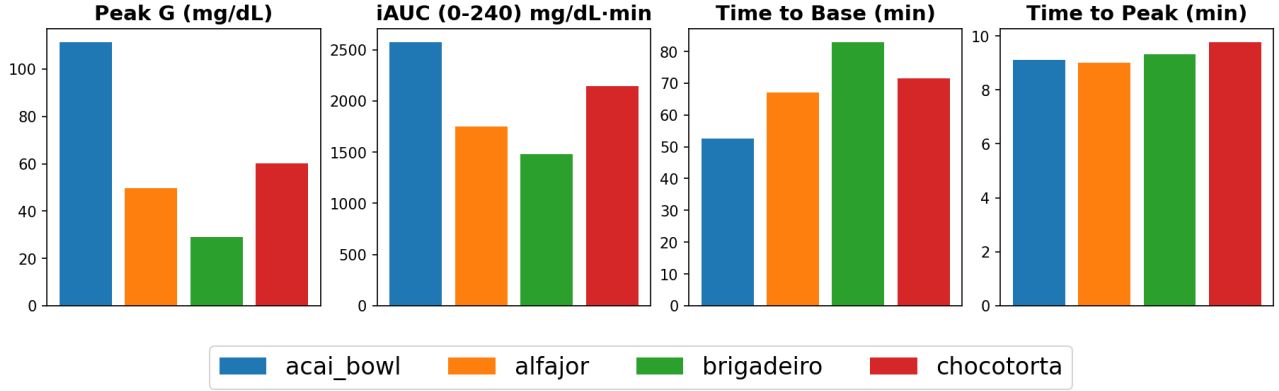


Figure 1: Side-by-side comparison of peak glucose, iAUC, and timing metrics across all desserts.

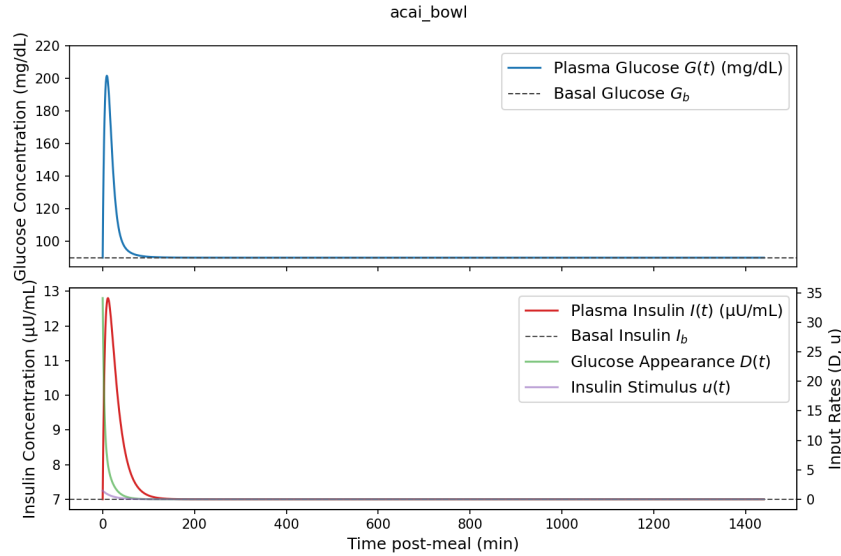


Figure 2: Full-day simulation (0 min to 1440 min) for the Açaí Bowl, showing return to baseline. Note: Input rates $D(t)$ (mg/dL/min) and $u(t)$ (μ U/mL/min) are plotted on the secondary y-axis.

4.2 Comparative analysis

Table 1 summarizes the key metrics for all simulated desserts, based on the generated `tables/summary.csv`. The **Açaí Bowl** resulted in, by far, the highest peak ΔG (111.48 mg/dL) and largest iAUC (integral of the total glucose above baseline), which is consistent with its high carbohydrate load (around 120 g per portion). However, it also exhibited the fastest time to return to baseline (52.50 min); this quick

Table 1: Summary of glycemic simulation results.

Dessert	Peak ΔG (mg/dL)	t_{peak} (min)	iAUC ₀₋₁₂₀ (mg/dL·min)	iAUC ₀₋₂₄₀ (mg/dL·min)	Time to Baseline (min)	Insulin Peak Time (min)
Açaí Bowl	111.48	9.12	2567.93	2576.19	52.50	11.50
Alfajor	49.61	9.00	1722.33	1748.62	67.00	11.50
Brigadeiro	28.91	9.32	1408.25	1479.26	83.00	11.50
Chocotorta	60.20	9.78	2114.93	2145.26	71.50	11.50

time can be attributed to the strong insulin response triggered by both the high glucose load and the significant protein content (24 g). [10]

In stark contrast, the **Brigadeiro** had the lowest peak (28.91 mg/dL) but the slowest return to baseline (83.00 min). Despite having the lowest carbohydrate content, the lack of significant protein and fiber is likely the reason for a result with weaker and slower insulin response, delaying the clearance of glucose.

The **Chocotorta** and **Alfajor**, the two Argentine deserts made from similar ingredients, showed intermediate responses; the significant presence of fat and fiber in these desserts modulates the appearance rate, [12] preventing the sharp spikes seen with pure sugar solutions, but the specific macronutrient ratios still lead to distinct profiles.

Finally, it is observed that the insulin peak time remains constant at 11.50 min across all scenarios. This invariance validates the solver’s stability, confirming protein kinetics decouple from glucose feedback as expected. While the *magnitude* of the insulin response scales with protein content, the *temporal dynamics* of the peak are determined solely by the aforementioned global rate constants and remain invariant in this simulation framework.

4.3 Nutrient sensitivity

A separate sensitivity analysis perturbed fat, fiber, and protein contents by $\pm 20\%$ for each dessert. The resulting metrics, written to `tables/sensitivity.csv` (see Table 2), confirm the expected qualitative behaviour suggested by the literature: **increasing fiber and fat generally lowers and delays the glucose peak by slowing gastric emptying and absorption**, [8, 14, 12] **whereas increasing protein mainly boosts the insulin response with only a modest direct effect on peak glucose**. These sensitivity findings support the internal consistency of the appearance model and illustrate how the simulator can be used to explore “what-if” reformulations of any given dessert.

5 Conclusion, limitations, and outlook

To conclude, the glycemic impact of four desserts using a nutrition-label-driven Bergman minimal model have successfully been simulated through this project’s pipeline. In more detail, the results highlight glycemic response not being solely determined by carbohydrate content but, in fact, significantly modulated by protein, fat, and fiber. The Açaí Bowl, despite a massive carbohydrate load, cleared relatively quickly due to a robust insulin response, whereas the low-protein Brigadeiro showed a prolonged return to baseline.

Moreover, the observed magnitudes of glucose excursions are within physiologically reasonable ranges for healthy individuals consuming such loads. However, the model does not come without its limitations. For one, it uses a single-compartment assumption for the remote insulin effect and relies on a nominal parameter set that does not account for inter-individual variability in insulin sensitivity or beta-cell function. [15] In layman’s terms this means the model simulates a generic, average person, ignoring unique individual biological differences. To add onto this simplification, the gut appearance mapping is also a simplification of complex gastrointestinal processes. [8, 11]

Lastly, from an application standpoint, an important limitation of this particular study is the relatively narrow set of desserts considered. All four items are high-sugar, moderately fatty South American desserts with limited fiber, all factors leading to broadly similar response shapes. A more diverse panel, including fiber-rich or lower-carbohydrate desserts, would likely produce more varied trajectories and sharper contrasts in the summary metrics. This limitation is, however, largely methodological rather than structural: because inputs are supplied via frozen YAML files, the client can readily add or replace desserts with any product for which a nutrition label is available, without having to modifying the code.

Future work could involve expanding the model to handle multiple sequential meals, adding stochastic variability, and incorporating more detailed gastric emptying dynamics. These additions would further enhance its predictive power and realism.

Authorship statement

This project was developed with assistance from AI tools. OpenAI’s ChatGPT / ChatGPT Codex was used during development to propose code structure, suggest implementations for some of the more complex helper functions, refactor infrastructure code (e.g. metadata handling, error hooks), and provide suggestions for how to present code in a cleaner fashion or simplify functions; Google’s “Jules” (Gemini 3.0) agent was used primarily for drafting and polishing documentation.

While AI(s) generated infrastructure, most core physiological logic was manually implemented and audited line-by-line against the scientific literature to prevent hallucination and ensure correction. AI was consulted for the manual implementation, providing a useful teaching role.

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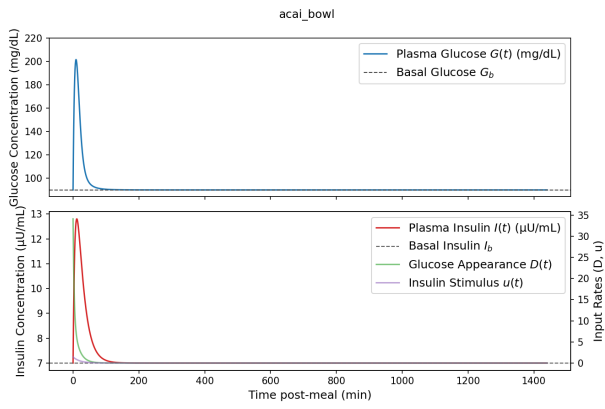
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Appendix

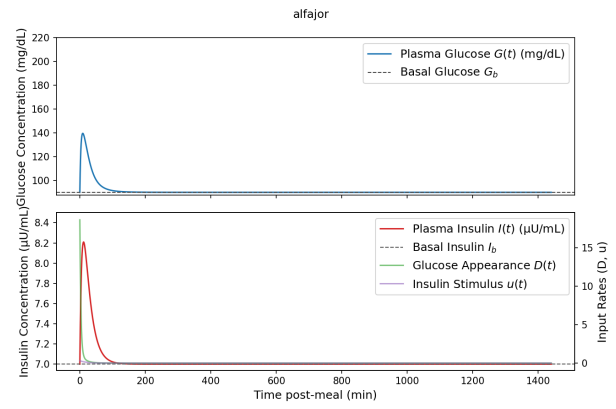
Table 2 presents the complete numerical results from the sensitivity analysis, where fat, fiber, and protein contents were perturbed by $\pm 20\%$.

Table 2: Detailed metrics from the sensitivity analysis sweep.

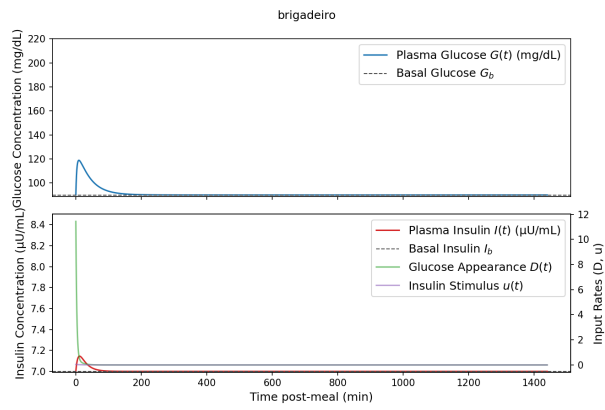
Dessert	Nutrient	Factor (multiplier)	Peak ΔG (mg/dL)	t_{peak} (min)	iAUC ₀₋₂₄₀ (mg/dL·min)
Açaí Bowl	Fat	0.8	112.45	9.08	2582.72
Açaí Bowl	Fat	1.2	110.54	9.16	2569.92
Açaí Bowl	Fiber	0.8	111.48	9.12	2576.19
Açaí Bowl	Fiber	1.2	111.48	9.12	2576.19
Açaí Bowl	Protein	0.8	114.40	9.68	2910.27
Açaí Bowl	Protein	1.2	109.02	8.68	2330.48
Alfajor	Fat	0.8	49.75	8.94	1749.33
Alfajor	Fat	1.2	49.47	9.06	1747.93
Alfajor	Fiber	0.8	49.94	8.99	1759.94
Alfajor	Fiber	1.2	49.27	9.01	1737.33
Alfajor	Protein	0.8	49.87	9.19	1893.80
Alfajor	Protein	1.2	49.36	8.82	1628.44
Brigadeiro	Fat	0.8	28.91	9.32	1479.27
Brigadeiro	Fat	1.2	28.90	9.33	1479.26
Brigadeiro	Fiber	0.8	28.96	9.32	1481.97
Brigadeiro	Fiber	1.2	28.85	9.32	1476.56
Brigadeiro	Protein	0.8	28.93	9.36	1502.08
Brigadeiro	Protein	1.2	28.89	9.29	1457.12
Chocotorta	Fat	0.8	60.50	9.70	2146.57
Chocotorta	Fat	1.2	59.91	9.86	2144.01
Chocotorta	Fiber	0.8	60.80	9.77	2165.29
Chocotorta	Fiber	1.2	59.61	9.79	2125.32
Chocotorta	Protein	0.8	60.68	10.08	2344.79
Chocotorta	Protein	1.2	59.76	9.54	1983.64



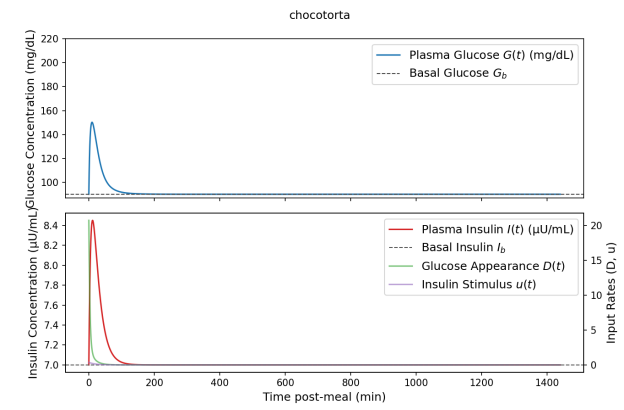
(a) Açaí Bowl



(b) Alfajor



(c) Brigadeiro



(d) Chocotorta

Figure 3: Postprandial glucose and insulin dynamics (0 min to 240 min) for all four desserts. Note: Input rates $D(t)$ (mg/dL/min) and $u(t)$ (μU/mL/min) are plotted on the secondary y-axis.